

AI operating system

Ten diagrams, each proving a different claim about the same system. Six architecture changes landed in the three weeks before this revision; four plates were redrawn and one is new. Prepared as reference artwork for the public ai-systems repository.

10	05	04	September 2026
plates	redrawn or added	figures contested	compiled

DERIVED FROM

ai-systems-brief.md — *positioning, visual grammar, corrections table*

skill-ecosystem-note.md — *architecture inventory and counts*

resume-bullets — *twelve named engineering mechanisms*

the September repository read — *six dated upgrades, 23 Aug to 13 Sep*

Every figure in this set carries its own evidence tier. There is no blanket disclaimer, because a single global caveat tells a reader that something is unverified without telling them which thing, and nothing on any page can then be trusted. The tiers cost more to maintain and are worth it.

Visual grammar

The plates form one system. Colour carries meaning and each colour means exactly one thing; form carries meaning independently of colour. Where a distinction matters, it is drawn as topology rather than hue, so it survives a monochrome print.

COLOUR

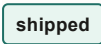
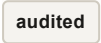
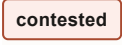
 Teal	Platform substrate: layers, stages, checks, lanes
 Purple	Products, and nothing else
 Coral	Enforcement: gates, halts, drift-prone surfaces
 Gray	Structural: frames, origins, inputs, outputs

FORM

Dashed outline	A gate or boundary condition. Never decorative.
Unit ticks	A count of agents, stages, rounds or attempts. Ticks reconcile by construction.
Numeral	A threshold. Drawn as a number so it cannot be mistaken for a reconciling count.
Direction	On plate 05, outcome. Up and left returns, down terminates, straight continues.
Taper	Distillation. Narrower layers sit above wider ones.
Broken rhythm	A deliberate exception. Plate 08's human gate is the case.

Evidence tiers

Each plate states what its claim rests on. Nothing is tiered higher than its weakest input: a plate whose structure is shipped but whose count is audited is labelled for both.

	A named file or package in the repository implements it. Ask to see the file.
	Counted or observed during a repository read, or confirmed by the author without a named file. Not re-counted from code.
	Two sources give different values and the conflict is unresolved. Treat as unknown, not as an average.
	No traceable source. Disregard until checked.

The ten plates

Each plate proves one claim. Where two would prove the same thing, one of them is wrong — so the set is not ten views of the architecture, it is ten separate arguments about it.

01	The operating system	A system with a build side and a run side, not a folder of scripts.
02	The knowledge substrate	A citation graph whose important nodes are computed, not nominated.
03	The compiler spine	Orchestration with isolation, not a prompt chain.
04	The contract spine	One source of truth, and the impact of changing it is computed.
05	The verification stack	Four paths, and the evidence is frozen before any of them is chosen.
06	Context residency	Budgeting is a measurement, and so is uptake.
07	The hardening loop	Promotion requires a proof of convergence.
08	The learning loops	Two loops: one a person gates, one that earns its own entries.
09	The runtime engine	Production operation with cost governance, not just a build pipeline.
10	Two-lane execution	The constraint is scoped, not total.

Corrections and additions

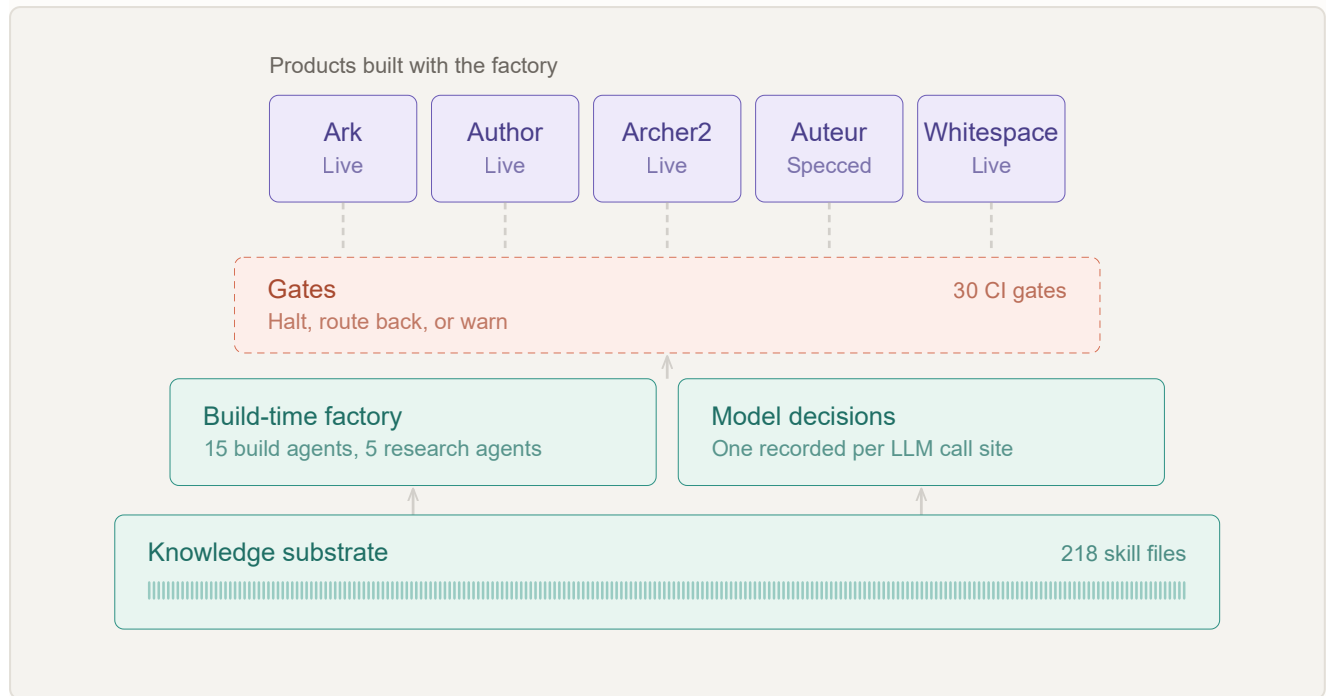
The previous set was compiled before six architecture changes landed between 23 August and 13 September. Most of these rows are additions. Two are corrections to claims the set was actively making: the verification plate drew two outcomes for a fortnight after the system grew a third, and it began the stack at the first check when the stack in fact begins one step earlier. The last two rows are neither: they are figures that were stated with more confidence than the sources support, and they have been demoted rather than quietly picked.

FIGURE	WAS	NOW	FROM
Plates	9	10	protocol
Verification outcomes	halt or warn	pass, warn, route back, halt	code
Where the stack begins	at the first check	at the evidence commitment	code
Knowledge substrate	8 named layers	a citation graph, 11 derived hubs	code
Context claim	~70% fewer tokens	tokens given against tokens used	code
Learning	one human-gated loop	two loops, one of them automatic	code
Products	four	five, one at a public URL	code
Exploration	absent	plate 10	audit
Preflight gates	19	19 or 30, unresolved	contested
Architecture layers	8	8 or 9, unresolved	contested

Rows sourced from *code* rest on named files. Rows marked *contested* are the ones worth dwelling on: two sources disagree, and the set draws the narrower claim with the conflict on the face of the plate rather than averaging them or choosing the more impressive. A reader who finds one undeclared conflict stops trusting every other number, which is why declaring them costs less than hiding them.

The operating system

A system with a build side and a run side, not a folder of scripts.



READING

The taper is the argument: one substrate carries a build side and a run side. Narrower layers sit above wider ones, so the compression from the corpus to the products is visible rather than asserted.

CHANGED FROM THE PREVIOUS SET

A fifth product joined the band. Whitespace Hunter went live on 2 September at a public URL, which makes it the only claim on this plate a reader can settle without trusting the plate.

EVIDENCE

audited agent counts

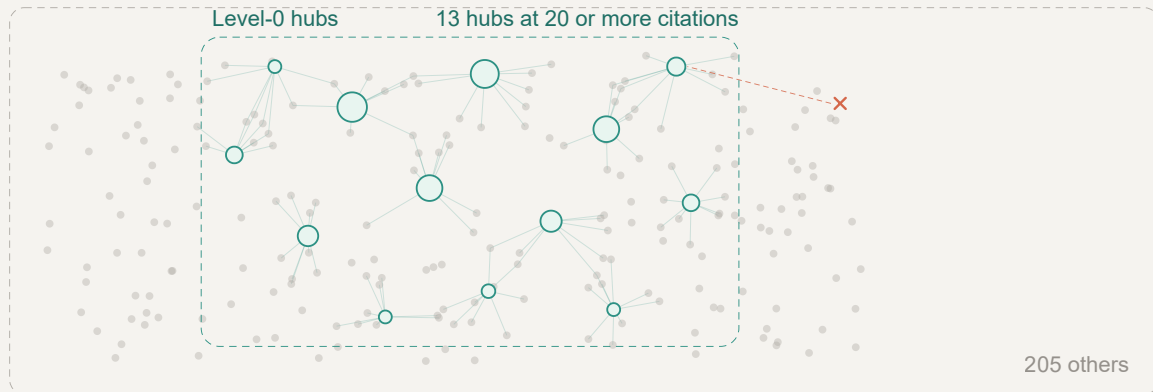
contested 152 files, gate count

unverified product status

The knowledge substrate

A citation graph whose important nodes are computed, not nominated.

218 skill files. The hubs are measured from the citation graph.



Derived, then enforced

skill-ref-count.ts --check-level0

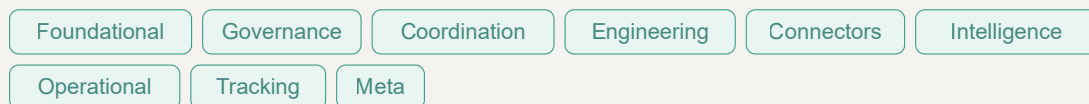
Counts citations between skill files and fails a commit when CLAUDE.md stops naming a measured hub

Wired to .husky/pre-commit, 12 Sep

The defect it was built for

agent-00 cited a section anchor that did not resolve. The spec validator was handed [section not found] where its constitution should have been. Fixed 12 Sep.

Nine layers group the graph; they do not rank it



READING

A skill file becomes Level-0 by being cited twenty times by other skill files. Nobody nominates it, and a pre-commit hook gates on the set the graph produces. Hubs are drawn at equal size because the ranking exists but the per-hub counts were never recorded.

CHANGED FROM THE PREVIOUS SET

Redrawn, not corrected. The plate stacked eight named layers, which is a filing cabinet: it said the corpus was organised, the least interesting true thing about it. The taxonomy survives as background; the graph is now the subject.

EVIDENCE

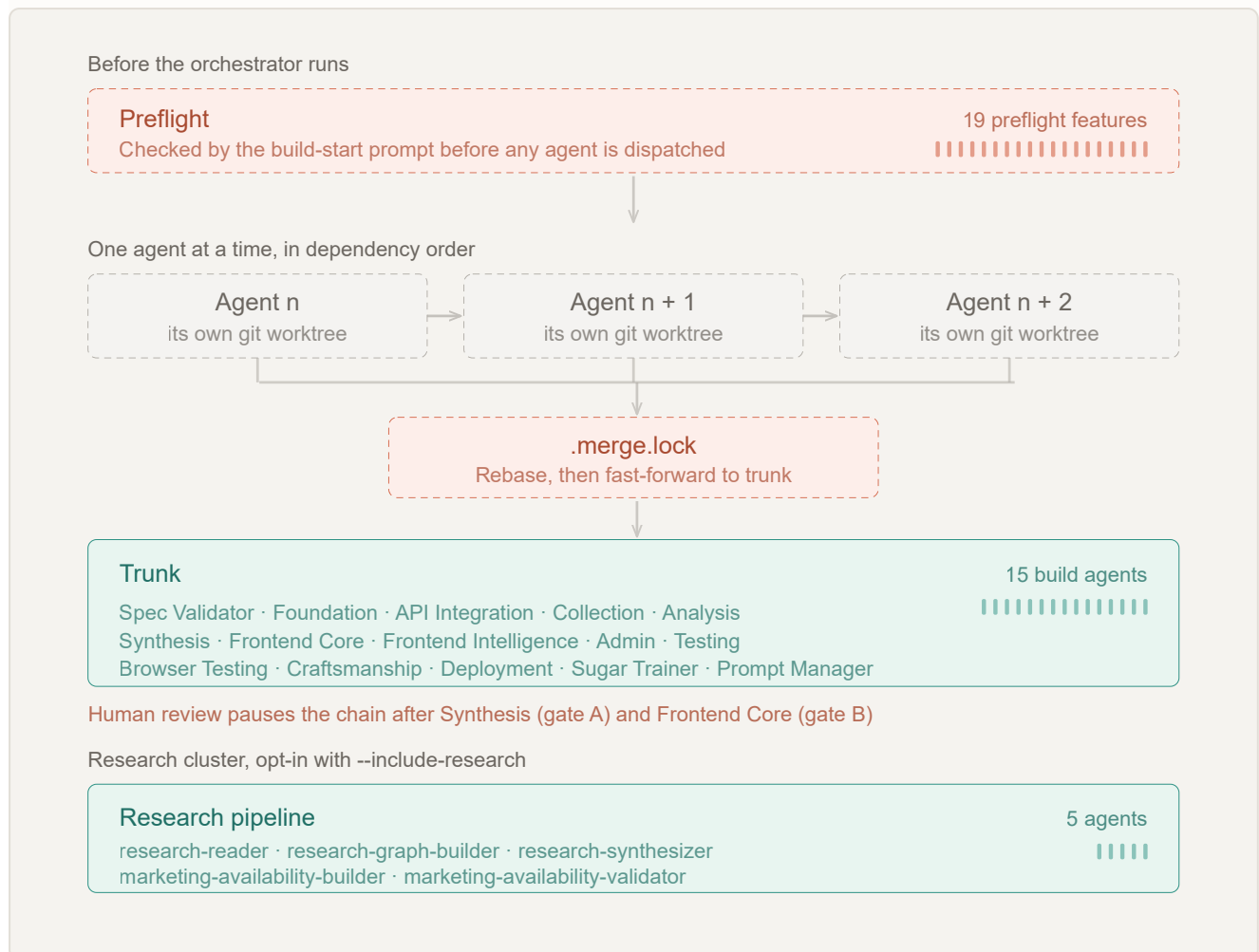
shipped skill-ref-count.ts, .husky/pre-commit

audited 11 hubs

contested 152 files, 8 or 9 layers

The compiler spine

Orchestration with isolation, not a prompt chain.



READING

Agents run concurrently in separate git worktrees and contend for the trunk through a file-lock mutex. Drawn as a linear chain, this was wrong.

CHANGED FROM THE PREVIOUS SET

Only the gate label moved. feature-descriptor.md reports nineteen preflight gates and verify-all.ts reports thirty checks; whether that is a superset or a different object is unresolved, so the plate draws the narrower claim and says why.

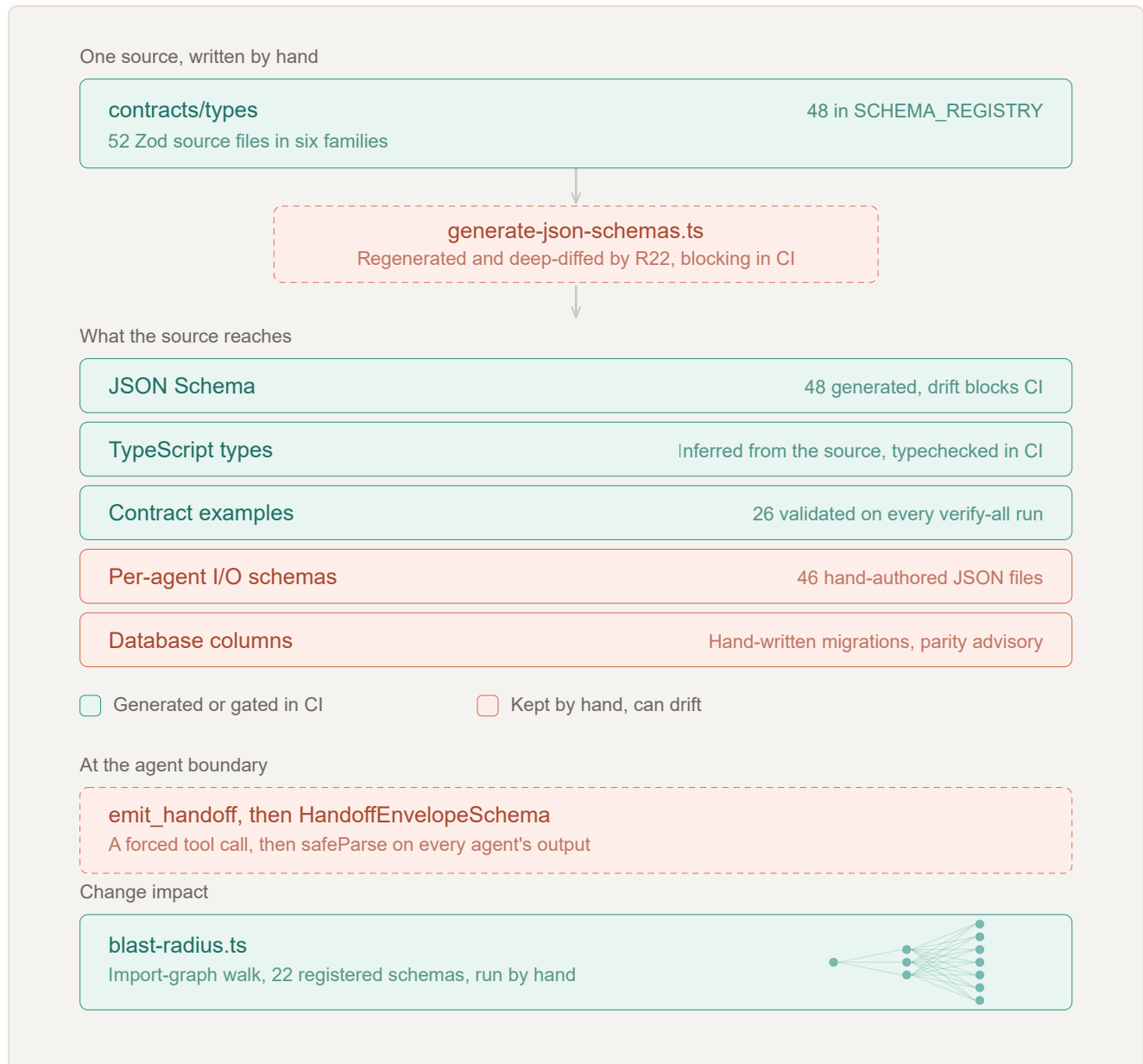
EVIDENCE

shipped master-agentic-orchestrator.ts

contested 19 against 30

The contract spine

One source of truth, and the impact of changing it is computed.



READING

A single schema derives every downstream surface. Generated surfaces and hand-maintained ones are drawn at equal prominence: the drift is labelled rather than airbrushed.

CHANGED FROM THE PREVIOUS SET

Nineteen hand-authored contracts are reported to derive forty-three generated schemas. If that holds it is the compression argument as a number, so it is drawn as a label rather than as structure until the two counts are reconciled.

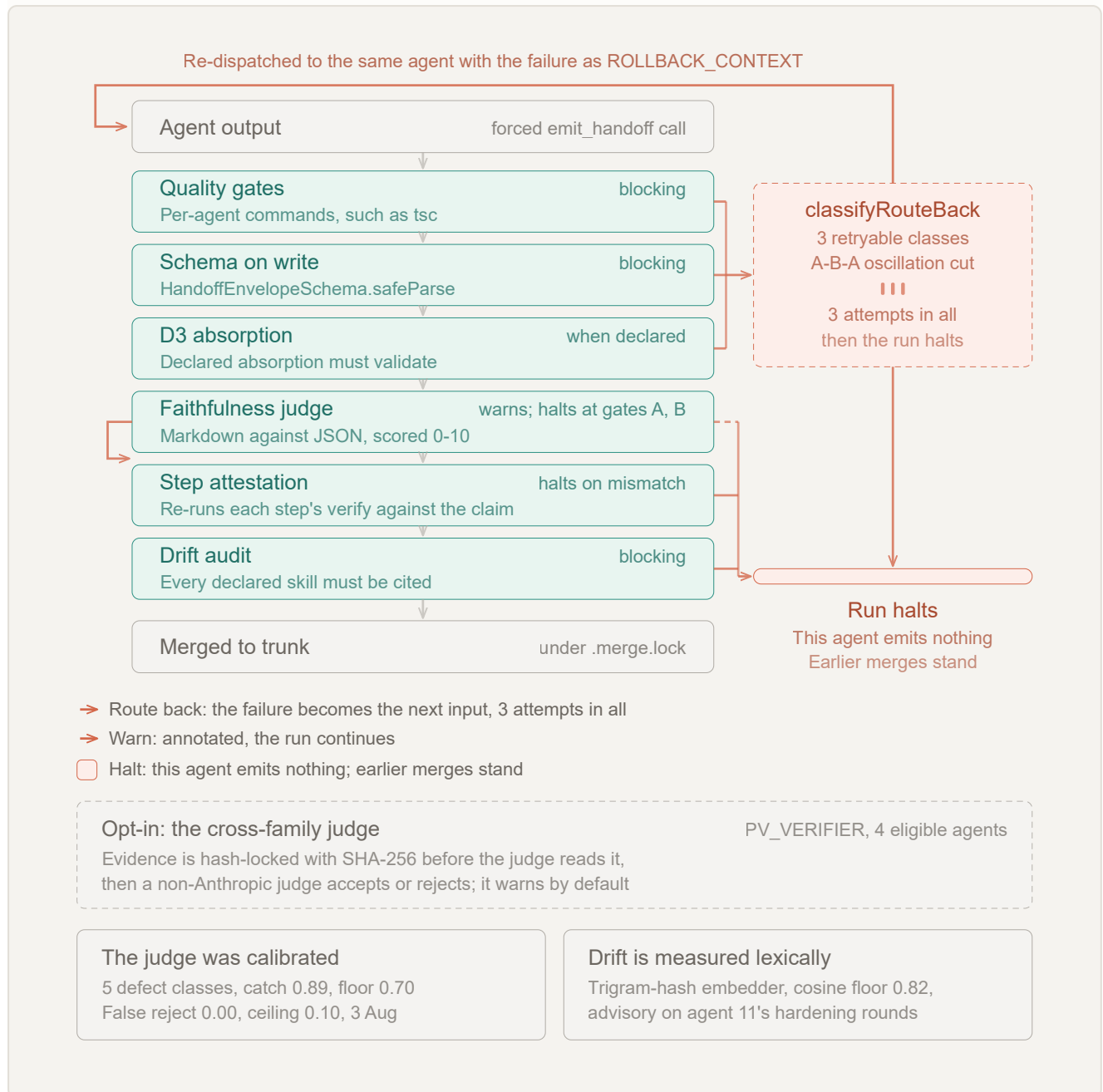
EVIDENCE

shipped contract-compiler.ts, blast-radius.ts

contested 19 against 43

The verification stack

Four paths, and the evidence is frozen before any of them is chosen.



READING

Direction encodes outcome: up and to the left returns, down terminates, straight continues. A failure is classified before it is acted on, so a compiler error is re-dispatched with its traceback as input while a fabrication halts. The attempt cap is drawn because a retry loop without a visible ceiling is the thing an experienced reader is already worried about.

CHANGED FROM THE PREVIOUS SET

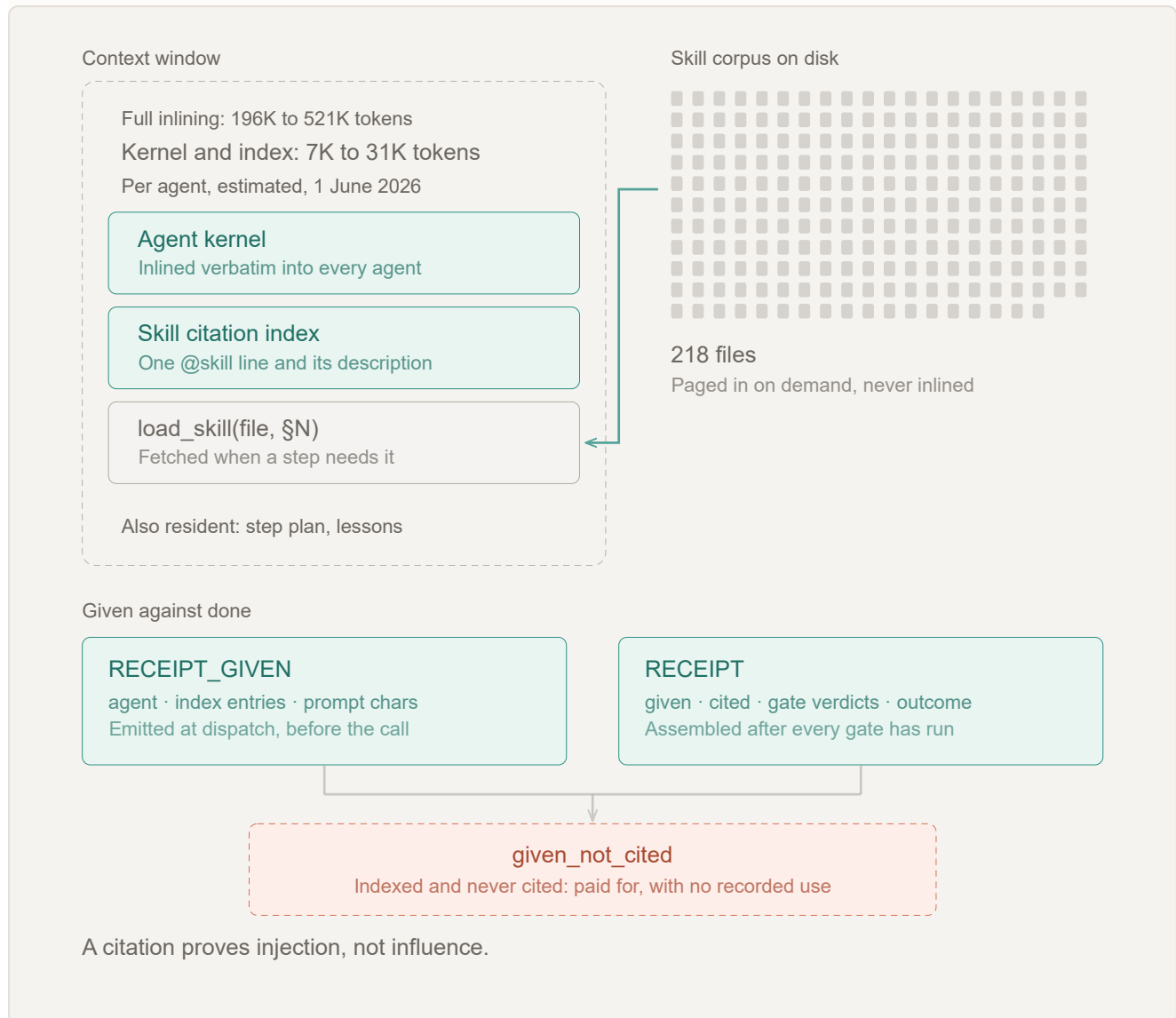
The most consequential correction in this revision. The plate drew two outcomes for the fortnight after the system grew a third. It also began at the first check, when the stack actually begins one step earlier: evidence is hash-locked before anything reads it, so the judge cannot be accused of grading a moving target.

EVIDENCE

shipped EvidenceCommitmentSchema, classifyRouteBack,
residual-baseline.ts

Context residency

Budgeting is a measurement, and so is uptake.



READING

What sits inside the window and what waits outside it. Token reduction is a cost claim; receipts make the harder one, recording what was injected against what the model actually cited. The industry treats those as the same event.

CHANGED FROM THE PREVIOUS SET

The efficiency figure is now half the plate rather than all of it. The two receipt records are drawn as structure rather than as bars, because no measured series has been recorded and drawing one would be inventing it.

EVIDENCE

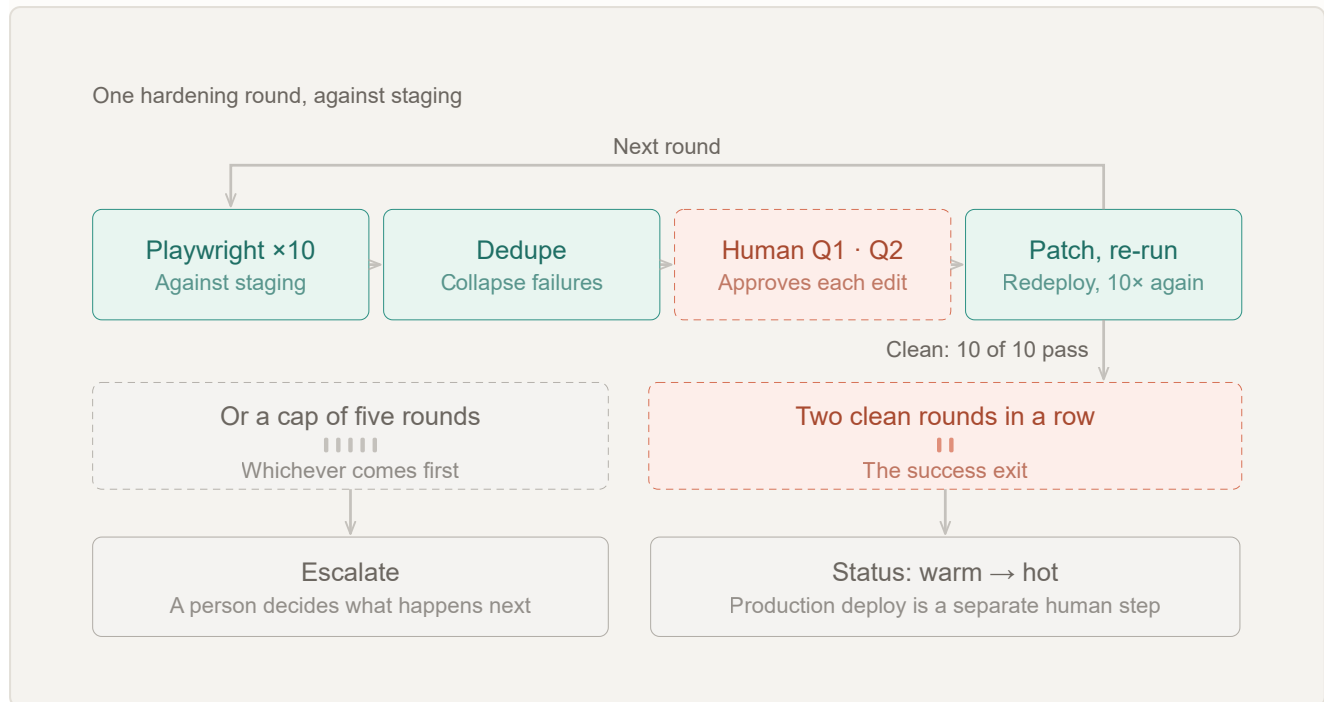
shipped PROGRESSIVE_DISCLOSURE, receipt.ts

audited ~70% reduction

unverified class counts

The hardening loop

Promotion requires a proof of convergence.



READING

The loop has an exit condition rather than an operator's judgement: two clean rounds in a row, or it keeps going. The human sits inside the loop as a steering call, not above it as an approver.

CHANGED FROM THE PREVIOUS SET

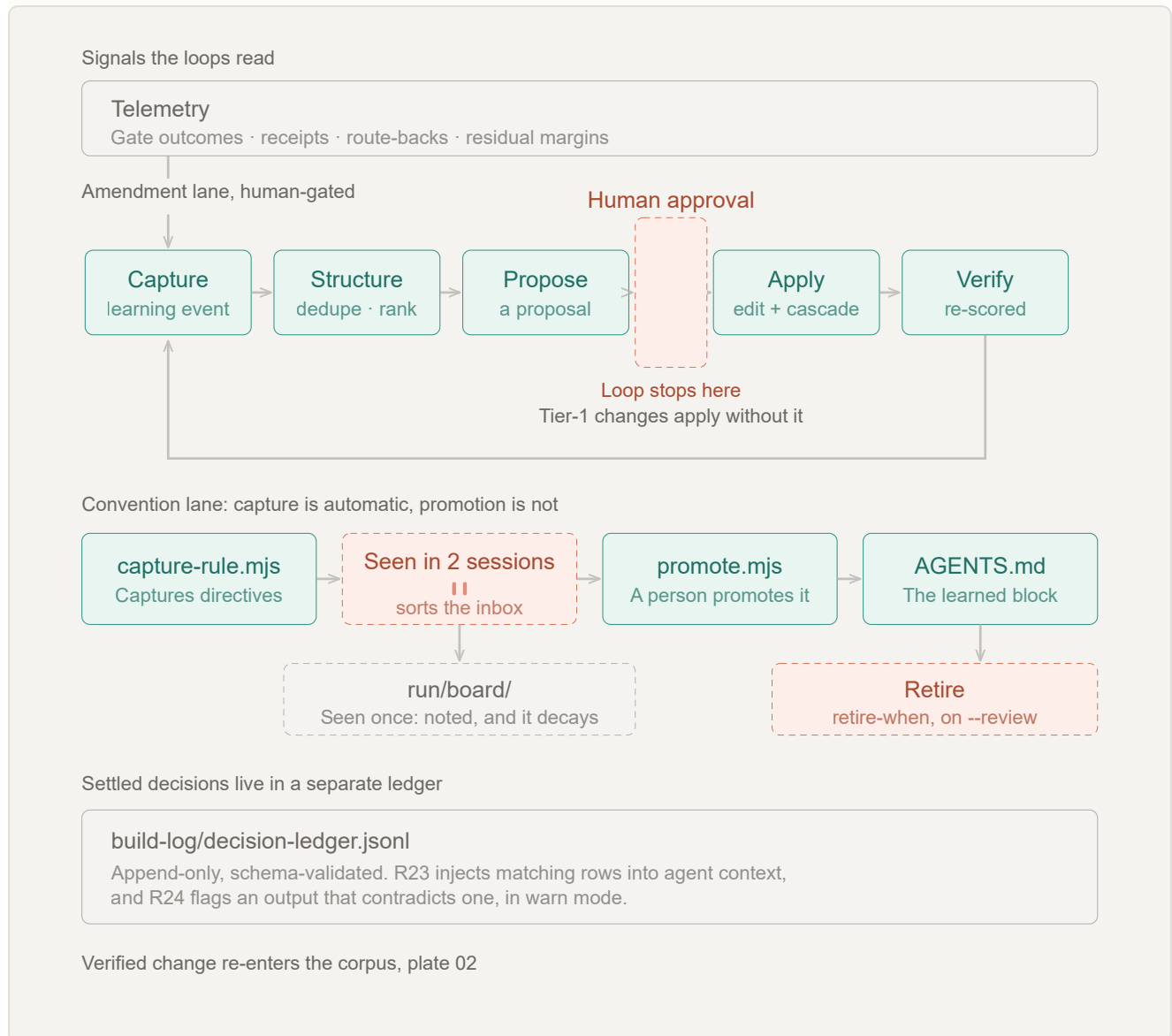
Unchanged but for its ticks. The termination condition now draws its two rounds as unit marks, so the count and the claim cannot drift apart.

EVIDENCE

`shipped agent-11-sugar-trainer.config.ts`

The learning loops

Two loops: one a person gates, one that earns its own entries.



READING

A rule reaches permanent context only after it has been observed in two independent sessions, and it leaves again when its retire condition fires. A memory that only grows is a memory nobody trusts, so the decay rule is drawn rather than described.

CHANGED FROM THE PREVIOUS SET

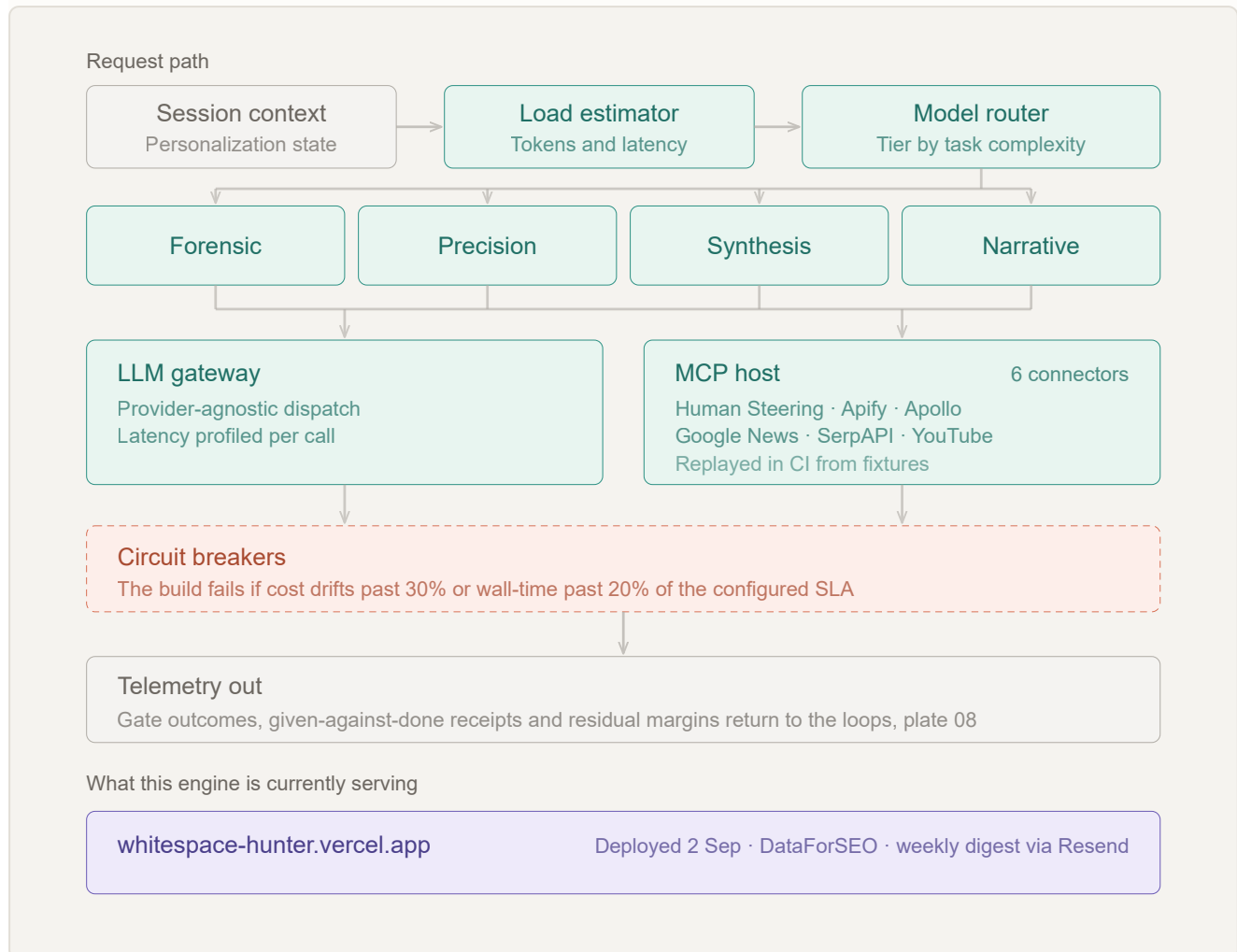
Promoted from designed protocol to shipped code, which is the most valuable kind of change in this corpus. The convention lane, the promotion threshold and the decay rule are files, not intentions, and the ledger both lanes write to is append-only.

EVIDENCE

shipped capture-rule.mjs, promote.mjs, AGENTS.md, decision-ledger.jsonl

The runtime engine

Production operation with cost governance, not just a build pipeline.



READING

Routing is a decision the system makes per task, and cost is a condition that can stop a run, so both belong in the same grammar as the build-time gates.

CHANGED FROM THE PREVIOUS SET

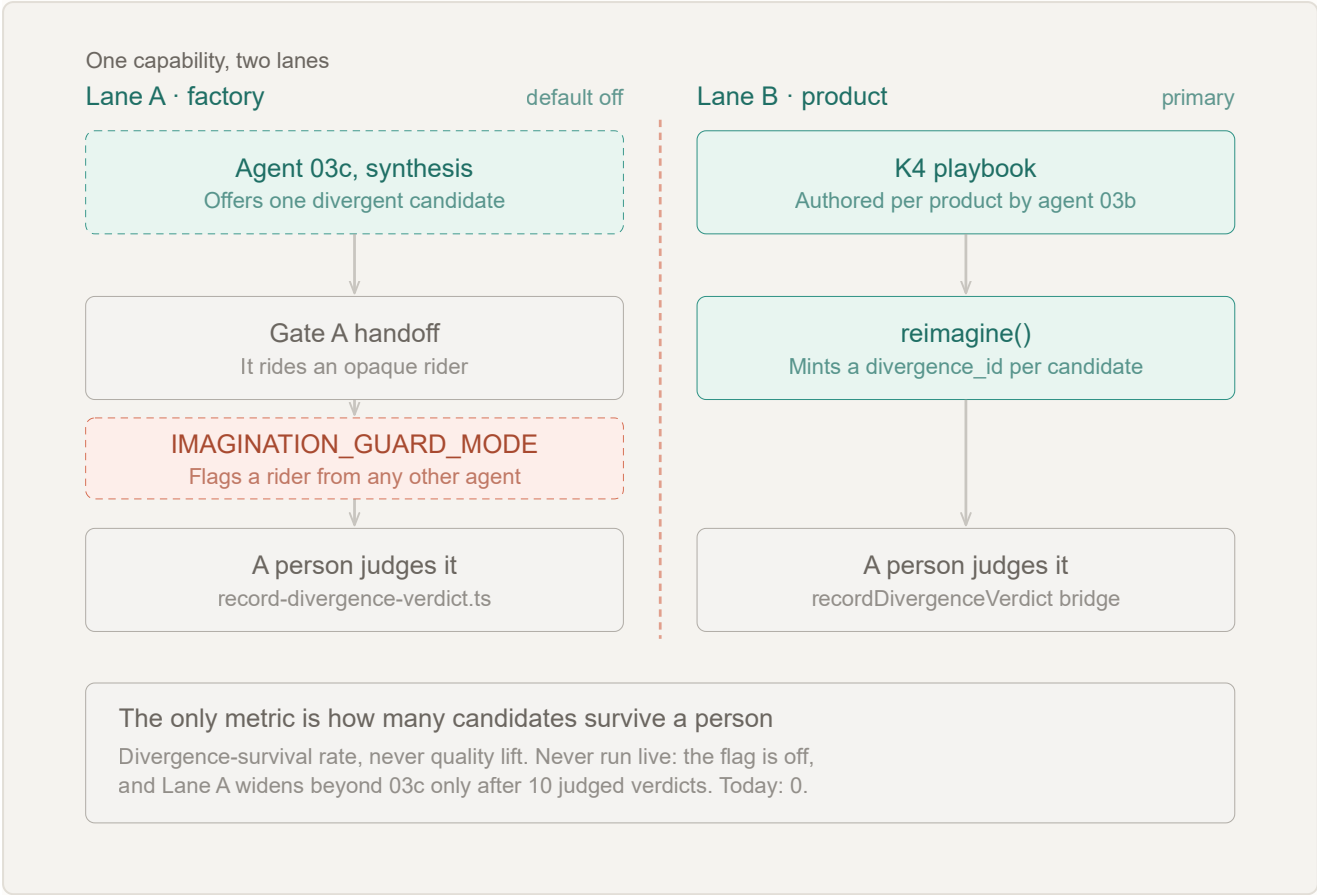
The telemetry that leaves this engine now includes receipts and residual margins, and the last row names a running service. A URL either loads or it does not, which makes it the cheapest evidence in the set.

EVIDENCE

shipped model-router, llm-gateway, mcp-infra, whitespace-hunter

Two-lane execution

The constraint is scoped, not total.



READING

Lane 2 is drawn dashed throughout because it is entirely boundary condition, and the boundary has exactly one opening. What crosses is an identifier under a parity check, never a payload, so the guarantee is visible as the absence of a second crossing rather than as a labelled box.

CHANGED FROM THE PREVIOUS SET

New. Every other plate argues that the system refuses things, which leaves an unanswered objection: a pipeline this rigid cannot produce anything new. This is also the least evidenced plate in the set, and its caption says so.

EVIDENCE

audited imagination protocol v1.1, no implementing file named

Structure

The counts that carry the structural argument. None has been re-counted from code, and six are actively contested between two sources. Author confirmation raises a *mechanism* to shipped when the source names the file that implements it, because a file is a thing a reader can be shown. It does not raise a *count*, because a count has no file to point at — only an arithmetic nobody has re-run.

FIGURE	VALUE	TIER	SOURCE	CHANGED
Skill files	152, possibly 164	contested	note; Stage 7 wired 12 parked skills	was 204
Architecture layers	8 or 9	contested	note gives 8; the September read gives 9	was 9, unnamed
Preflight gates	19	contested	feature-descriptor.md	was 21
Verification checks	30	contested	verify-all.ts; relationship to the 19 unknown	absent
Typed handoff contracts	19	contested	contracts/CONTRACT_CATALOG.md	absent
Generated JSON schemas	43	contested	unified Zod tree	absent
Contract families	3	audited	note	absent
Build agents	15	audited	note, corroborated twice	was six stages summing to 20
Research agents	5, named	audited	note, corroborated twice	absent
Total agents	20	audited	two independent sources agree	the one figure drawn bare
Level-0 hub skills	11	audited	skills at 20 or more citations	absent
Graded prompt corpus	471	audited	residual baseline corpus	absent

Mechanism

Every mechanism drawn in the set, the file that implements it, and when it landed. The dates are part of the evidence: a corpus showing eleven dated changes in three weeks is a system under active development, which is a different and better claim than a finished portfolio piece.

FIGURE	VALUE	SOURCE	LANDED
Pipeline topology	Parallel, git-worktree isolated, mutex-gated merge	master-agentic-orchestrator.ts	standing
Merge serialisation	.sandboxes/.merge.lock	master-agentic-orchestrator.ts	standing
Contract enforcement	Structural subtyping asserted at build time	contract-compiler.ts	standing
Change impact	Breadth-first traversal of the contract import graph	blast-radius.ts	standing
Runtime prompt surface	Compiled from contracts, not hand-written	agent-12-prompt-manager.config.ts	standing
Decode discipline	Constrained decoding via forced emit_handoff	master-agentic-orchestrator.ts	standing
Faithfulness judge	Markdown against JSON, temperature 0.2, scored 0-10	verify-handoff-faithfulness.ts	standing
Drift embedder	trigram-hash-64-v1, deterministic, zero cost	detect-semantic-drift.ts	standing
Drift threshold	0.82 cosine against baseline	detect-semantic-drift.ts	standing
Enforcement ladder	disabled, advisory, enforcing	detect-semantic-drift.ts	standing
Judge catch rate floor	At least 0.70	measure-judge-catch-rate.ts	standing
Judge false reject ceiling	At most 0.10	measure-judge-catch-rate.ts	standing
Context reduction	Roughly 70% of system prompt tokens	PROGRESSIVE_DISCLOSURE	standing
Hardening round size	10 Playwright runs per round	agent-11-sugar-trainer.config.ts	standing
Termination condition	Two consecutive clean rounds	agent-11-sugar-trainer.config.ts	standing
Router tiers	Forensic, Precision, Synthesis, Narrative	packages/model-router	standing
Cost circuit breaker	Cost drift past 30% of SLA	packages/llm-gateway	standing
Wall-time circuit breaker	Wall-time drift past 20% of SLA	packages/llm-gateway	standing
MCP connectors	6, named	packages/mcp-infra	standing
Market discovery service	Live, public, DataForSEO-backed	whitespace-hunter.vercel.app	1-2 Sep
Decision ledger	Append-only JSONL, schema-validated	build-log/decision-ledger.jsonl	1-6 Sep
Evidence commitment	SHA-256 hash-lock taken before the judge reads anything	EvidenceCommitmentSchema	1-6 Sep
Route-back classifier	Failure traceback re-dispatched as structured input	classifyRouteBack	1-6 Sep

FIGURE	VALUE	SOURCE	LANDED
Route-back cap	At most 3 attempts, A-B-A oscillation detected	classifyRouteBack	1-6 Sep
Given-vs-done receipts	RECEIPT_GIVEN against RECEIPT	receipt.ts	1-6 Sep
Residual baseline	score, floor and margin tracked over time	residual-baseline.ts	1-6 Sep
Two-lane execution	Lane 1 strict, Lane 2 exploratory, one crossing	imagination protocol v1.1	6-13 Sep
Divergence transport	divergence_id under K4 parity check	agent 03c explore mode	6-13 Sep
Level-0 hub gate	Commits gated on hub-skill integrity, 20 citations	skill-ref-count.ts, .husky/pre-commit	12 Sep
Convention flywheel	Rule promoted after 2 independent sessions	capture-rule.mjs, promote.mjs, AGENTS.md	12 Sep
Rule decay	Declarative stale-when and retire-when	convention lane	12 Sep

Unverified, and therefore not drawn

FIGURE	PREVIOUS VALUE	WHY IT IS OUT
Residency class counts	12 resident / 60 index / 132 paged	Sums to 204, which was the wrong total. The three classes survive; the counts do not.
Product status	Ark live, Author live, Archer2 dormant, Auteur spected	Not corroborated by any source document. Drawn, but plate 01 says so.
Per-layer file counts	never stated	No source gives a distribution. Plate 02 draws the corpus as a graph rather than as equal rows, so it implies none.
Build-time compression	2-4 weeks against 6-9 months	A comparative claim with no measured baseline on either side.
Intervention reduction	up to 80% fewer manual interventions	No instrument produces this number. The phrase 'up to' is doing all the work.
Trend surge magnitudes	18x, +1800%	Real outputs of a real tool, presented as headline figures rather than as one run's result. If used at all they belong to a named query on a named date.

Still owed to the repository

Contested is the weak rank in this set and audited is the next weakest. Twelve figures carry the structural argument and none has been re-counted from code. Record the command beside each number, not just the number: a figure with the command that produced it is evidence, and a figure without one is a claim.

- Count `SKILL.md` under the skills tree, against **152** or **164**, and state which definition was used
- Read the layer taxonomy in the skill-file standard: is `connectors` a ninth layer, against **8** or **9**
- Count gate definitions in the pipeline config, against **19**
- Count checks in `verify-all.ts`, against **30**, and settle whether they are a superset of the 19
- Count entries in `contracts/CONTRACT_CATALOG.md`, against **19**
- Count generated JSON schemas, against **43**, and confirm they derive from the 19
- Run `skill-ref-count.ts --check-level0` and record its output verbatim, against **11**
- Count the residual baseline corpus, against **471**
- Run the dual assembly and record what it actually prints, against **~70%**
- Name the file that implements two-lane execution — plate 10 is the only plate resting on a single unnamed source

One figure is documented but deliberately undrawn: the six-second type-check saving per dispatch from incremental `tsbuildinfo` seeding. No plate in this

set makes a claim it would support, and a figure without a claim is decoration.

One defect is recorded here because it is worth publishing rather than despite being embarrassing. `agent-00.config.ts` cited a section anchor that did not resolve, so the spec validator — first in the pipeline, the gate every build passes through — was handed the literal string `[section not found]` where its constitution should have been. A citation that silently resolves to an error string is worse than one that throws, because the pipeline keeps running and everything downstream is confidently built on nothing. Found by audit rather than by symptom, fixed on 12 September, and the Level-0 gate on plate 02 now catches the class.

The plates are compiled, not drawn. A grammar module holds the colour roles, ticks, gates and taper as primitives, and each plate is written in terms of those, so a grammar violation is impossible to express rather than something a reviewer has to catch. It is the argument plate 04 makes about schemas, applied to the artwork. Source and regeneration instructions ship beside the SVGs in `diagrams/_src`. Committed as SVG, not raster: it renders natively on GitHub, appears in diffs, and lets a reviewer watch the architecture change over time.